Smoke Generator

Category: **Plumbing & Pipe Fitting** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: Smoke Generator for testing of pipes and joints

What it is used for

A smoke generator introduces controlled smoke into a suitable pipe system to help identify leaks at joints and connections. Correct sealing, controlled smoke delivery, and suitable ventilation help make the test easier to interpret and safer to perform.

Main parts

- Smoke generation unit produces visible test smoke.
- Air pump or pressure source moves smoke through the test system.
- Smoke outlet hose connects the generator to the pipe system.
- Adapter or sealing plug closes the test connection.
- Flow or pressure control regulates smoke delivery on equipped models.
- Indicator or gauge shows operating status or pressure where provided.
- Power switch controls the unit on powered models.
- Protective casing encloses the generator components.

Before you start

1. Read the equipment manual and confirm the generator is suitable for the pipe material and test method.
2. Inspect the generator, hoses, adapters, seals, controls, and power supply for damage.
3. Identify the pipe section to be tested and isolate it from incompatible connected equipment.
4. Confirm the test system can safely withstand the specified smoke pressure.
5. Ensure adequate ventilation and clear the area of people not involved in testing.
6. Wear suitable PPE and check that the smoke fluid, if used, is approved for the device.

How to use

1. Switch off the generator before connecting hoses or adapters.
2. Seal the pipe openings using suitable adapters or plugs according to the test procedure.
3. Connect the smoke outlet hose securely to the designated test port.
4. Set the pressure or flow limit according to the pipe system and equipment instructions.
5. Switch on the generator and introduce smoke gradually into the isolated pipe section.
6. Observe accessible joints, seals, and connections for smoke escaping from unintended locations.
7. Mark suspected leak points and record the test conditions and observations.
8. Stop smoke delivery if the pressure limit is reached or abnormal conditions appear.
9. Switch off the generator, release residual pressure safely, and disconnect the hose only after pressure has returned to a safe level.

10. Remove test plugs, restore the pipe system as required, and inspect and store the equipment.

⚠ Safety

- Never exceed the pressure rating of the pipe system, seals, or generator.
- Do not use smoke testing on a live gas line or hazardous system unless an approved procedure specifically permits it.
- Keep smoke away from eyes and avoid inhalation; provide ventilation and follow the fluid safety instructions.
- Do not use the generator near flammable atmospheres or ignition sources unless the equipment is specifically approved for that environment.
- Never disconnect a pressurised hose or remove a sealing plug while pressure remains.
- Stop immediately if a hose bulges, a connection moves, or unexpected pressure develops.

Care & storage

- Switch off and disconnect power before cleaning or maintenance.
- Clean adapters and hoses according to the manufacturer's instructions and allow them to dry.
- Inspect hoses, seals, connectors, and casing for wear or damage after use.
- Store the generator and approved smoke fluid in a clean, dry area according to the manual.

Common mistakes to avoid

- Applying more pressure than the pipe system is rated to withstand.
- Leaving untested openings unsealed so smoke escapes before reaching the test section.
- Disconnecting hoses or plugs before releasing residual pressure.
- Using unapproved smoke fluid or testing in a poorly ventilated area.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.